

**Sandhya M 250701639-2025-CSE** ▾**S2****Started on** Tuesday, 11 August 2026, 11:02 AM**State** Finished**Completed on** Tuesday, 11 August 2026, 11:05 AM**Time taken** 3 mins 5 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

**Question 1** | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countZeroes(int a[], int low, int high)
4  {
5      if (low > high)
6          return 0;
7
8      if (a[low] == 0)
9          return high - low + 1;
10
11     int mid = (low + high) / 2;
12
13     return countZeroes(a, mid + 1, high);
14 }
15
16 int main()
17 {
18     int m;
19     scanf("%d", &m);
20
21     int a[m];
22
23     for (int i = 0; i < m; i++)
24         scanf("%d", &a[i]);
25
26     printf("%d", countZeroes(a, 0, m - 1));
27
28     return 0;
29 }
```

|   | Input                      | Expected | Got |   |
|---|----------------------------|----------|-----|---|
| ✓ | 5<br>1<br>1<br>1<br>0<br>0 | 2        | 2   | ✓ |

|   | Input                                                                                             | Expected | Got |   |
|---|---------------------------------------------------------------------------------------------------|----------|-----|---|
| ✓ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1                                    | 0        | 0   | ✓ |
| ✓ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0                                                    | 8        | 8   | ✓ |
| ✓ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2        | 2   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Sandhya M 250701639-2025-CSE ▾

S2

|              |                                   |
|--------------|-----------------------------------|
| Started on   | Tuesday, 11 August 2026, 11:05 AM |
| State        | Finished                          |
| Completed on | Tuesday, 11 August 2026, 11:06 AM |
| Time taken   | 1 min 13 secs                     |
| Marks        | 1.00/1.00                         |
| Grade        | 10.00 out of 10.00 (100%)         |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than  $\lfloor n / 2 \rfloor$  times. You may assume that the majority element always exists in the array.

**Example 1:**

Input: `nums = [3,2,3]`

Output: 3

**Example 2:**

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

**Constraints:**

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

**For example:**

| Input              | Result |
|--------------------|--------|
| 3<br>3 2 3         | 3      |
| 7<br>2 2 1 1 1 2 2 | 2      |

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int majorityElement(int nums[], int n)
4  {
5      int candidate = nums[0];
6      int count = 1;
7
8      for (int i = 1; i < n; i++)
9      {
10         if (nums[i] == candidate)
11             count++;
12         else
13             count--;
14
15         if (count == 0)
16         {
17             candidate = nums[i];
18             count = 1;
19         }
20     }
21
22     return candidate;
23 }
24
25 int main()
26 {
27     int n;
28     scanf("%d", &n);
29
30     int nums[n];
31
32     for (int i = 0; i < n; i++)
33         scanf("%d", &nums[i]);
34

```

```
35 |         printf("%d", majorityElement(nums, n));
36 |
37 |     return 0;
38 | }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 3     | 3        | 3   | ✓ |
|   | 3 2 3 |          |     |   |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



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S2

|              |                                   |
|--------------|-----------------------------------|
| Started on   | Tuesday, 11 August 2026, 11:06 AM |
| State        | Finished                          |
| Completed on | Tuesday, 11 August 2026, 12:07 PM |
| Time taken   | 1 hour                            |
| Marks        | 1.00/1.00                         |
| Grade        | 10.00 out of 10.00 (100%)         |

**Question 1** | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

**Output Format**

First Line Contains Integer – Floor value for x

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n, x;
6      scanf("%d", &n);
7
8      int a[n];
9
10     for (int i = 0; i < n; i++)
11         scanf("%d", &a[i]);
12
13     scanf("%d", &x);
14
15     int low = 0, high = n - 1;
16     int floor = -1;
17
18     while (low <= high)
19     {
20         int mid = (low + high) / 2;
21
22         if (a[mid] <= x)
23         {
24             floor = a[mid];
25             low = mid + 1;
26         }
27         else
28         {
29             high = mid - 1;
30         }
31     }
32
33     printf("%d", floor);
34
35     return 0;
36 }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 6     | 2        | 2   | ✓ |
|   | 1     |          |     |   |
|   | 2     |          |     |   |
|   | 8     |          |     |   |
|   | 10    |          |     |   |
|   | 12    |          |     |   |
|   | 19    |          |     |   |
|   | 5     |          |     |   |

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 5     | 85       | 85  | ✓ |
|   | 10    |          |     |   |
|   | 22    |          |     |   |
|   | 85    |          |     |   |
|   | 108   |          |     |   |
|   | 129   |          |     |   |
|   | 100   |          |     |   |
| ✓ | 7     | 9        | 9   | ✓ |
|   | 3     |          |     |   |
|   | 5     |          |     |   |
|   | 7     |          |     |   |
|   | 9     |          |     |   |
|   | 11    |          |     |   |
|   | 13    |          |     |   |
|   | 15    |          |     |   |
|   | 10    |          |     |   |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



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S2

|              |                                   |
|--------------|-----------------------------------|
| Started on   | Tuesday, 11 August 2026, 12:07 PM |
| State        | Finished                          |
| Completed on | Tuesday, 11 August 2026, 12:08 PM |
| Time taken   | 1 min 12 secs                     |
| Marks        | 1.00/1.00                         |
| Grade        | 10.00 out of 10.00 (100%)         |

**Question 1** | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findPair(int a[], int left, int right, int x, int *e1, int *e2)
4  {
5      if (left >= right)
6          return 0;
7
8      if (a[left] + a[right] == x)
9      {
10         *e1 = a[left];
11         *e2 = a[right];
12         return 1;
13     }
14
15     if (a[left] + a[right] < x)
16         return findPair(a, left + 1, right, x, e1, e2);
17
18     return findPair(a, left, right - 1, x, e1, e2);
19 }
20
21 int main()
22 {
23     int n, x;
24     scanf("%d", &n);
25
26     int a[n];
27
28     for (int i = 0; i < n; i++)
29         scanf("%d", &a[i]);
30
31     scanf("%d", &x);
32
33     int e1, e2;
34
35     if (findPair(a, 0, n - 1, x, &e1, &e2))
36     {
37         printf("%d\n", e1);
38         printf("%d\n", e2);
39     }
40     else
41     {
42         printf("No");
43     }
44
45     return 0;
46 }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 4     | 4        | 4   | ✓ |
|   | 2     | 10       | 10  |   |
|   | 4     |          |     |   |
|   | 8     |          |     |   |
|   | 10    |          |     |   |
|   | 14    |          |     |   |
| ✓ | 5     | No       | No  | ✓ |
|   | 2     |          |     |   |
|   | 4     |          |     |   |
|   | 6     |          |     |   |
|   | 8     |          |     |   |
|   | 10    |          |     |   |
|   | 100   |          |     |   |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Back to Course



Sandhya M 250701639-2025-CSE ▾

**S2****Started on** Tuesday, 11 August 2026, 12:09 PM**State** Finished**Completed on** Tuesday, 11 August 2026, 12:10 PM**Time taken** 47 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct | Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

| Input          | Result         |
|----------------|----------------|
| 5              | 12 34 67 78 98 |
| 67 34 12 98 78 |                |

Answer:

```

1  #include <stdio.h>
2
3  void quickSort(int a[], int low, int high)
4  {
5      if (low < high)
6      {
7          int i = low;
8          int j = high;
9          int pivot = a[low];
10
11         while (i < j)
12         {
13             while (a[i] <= pivot && i < high)
14                 i++;
15
16             while (a[j] > pivot)
17                 j--;
18
19             if (i < j)
20             {
21                 int temp = a[i];
22                 a[i] = a[j];
23                 a[j] = temp;
24             }
25         }
26
27         a[low] = a[j];
28         a[j] = pivot;
29
30         quickSort(a, low, j - 1);
31         quickSort(a, j + 1, high);
32     }
33 }
34 int main()
35 {
36     int n;
37     scanf("%d", &n);
38
39     int a[n];
40
41     for (int i = 0; i < n; i++)
42         scanf("%d", &a[i]);
43
44     quickSort(a, 0, n - 1);
45
46     for (int i = 0; i < n; i++)
47     {
48         printf("%d", a[i]);
49         if (i < n - 1)
50             printf(" ");
51     }
52

```

|   | Input                               | Expected                      | Got                           |   |
|---|-------------------------------------|-------------------------------|-------------------------------|---|
| ✓ | 5<br>67 34 12 98 78                 | 12 34 67 78 98                | 12 34 67 78 98                | ✓ |
| ✓ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✓ |
| ✓ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90    | 1 2 3 4 5 6 7 8 9 10 11 90    | 1 2 3 4 5 6 7 8 9 10 11 90    | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)